MIL-C-10664C
17 December 1981
SUPERSEDING
MIL-C-10664B
31 December 1958

MILITARY SPECIFICATION

CHLOROPICRIN, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of chloropicrin (CCl_3NO_2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "40 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if aplicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D891 - Specific Gravity of Industrial Aromatic Hydrocarbons and Related Materials
D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Appearance. Chloropicrin shall be a colorless liquid which is free from any clouds or droplets of separated moisture when tested as specified in 4.2.4.1.

3.2 Chemical and physical characteristics. Chloropicrin shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Minimum	Maximum	Test paragraph
Specific gravity at 25°/4°C	1.635		4.2.4.2
Residue on evaporation, percent by weight		0.50	4.2.4.3
Purity, percent by weight	98		4.2.4.4

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the chloropicrin produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. All containers shall be examined for packaging.

4.2.2.2 For test (see 6.4). Sampling shall be conducted in accordance with table II. A representative specimen of approximately 100 milliliters (ml) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

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TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. Filled unit and shipping containers shall be examined for the following defects. Containers having one or more defects shall be rejected.

(a) Container not as specified
(b) Container damaged or leaking
(c) Marking incorrect, missing, or illegible

4.2.3.2 For test (see 6.4). Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests (see 6.4). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Appearance. Visually examine the specimen for color and evidence of clouds or droplets of separated moisture.

4.2.4.2 Specific gravity. Determine the specific gravity of the specimen in accordance with the pycnometer method of ASTM D891 except weigh the pycnometer filled with water at 4°C rather than 15.56°C, and weigh the pycnometer filled with specimen at 25°C rather than 15.56°C.

4.2.4.3 Residue on evaporation. Add 5 to 10 ml of specimen to a weighing buret and weigh to the nearest 0.1 milligram. Transfer approximately 5 ml of the specimen into a small, tared evaporating dish. Reweigh the weighing buret. Evaporate to dryness on a steam bath in a well-ventilated hood. Dry in an oven at 105° $\pm$ 5°C for 1/2 hour, cool to room temperature in a desiccator, and weigh to the nearest 0.1 milligram. Calculate the percent by weight residue on evaporation as follows:

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$$\text{Percent residue on evaporation} = \frac{100A}{B}$$

where: A = Weight of residue in grams, and
B = Weight of specimen in grams.

4.2.4.4 Purity.

(a) Percent free chlorine. Pipet 25 ml of specimen into a glass-stoppered Erlenmeyer flask containing 75 ml of 5-percent potassium iodide solution. Shake thoroughly for at least 1 minute and titrate the liberated iodine with standardized 0.1N sodium thiosulfate solution using starch as the indicator. Calculate the percent by weight free chlorine as follows:

$$\text{Percent free chlorine} = \frac{0.142AB}{C}$$

where: A = Milliliters of sodium thiosulfate solution used,
B = Normality of the sodium thiosulfate solution, and
C = Specific gravity determined in 4.2.4.2.

(b) Procedure. Weigh to the nearest 0.1 milligram approximately 0.25 gram of the specimen into a blown glass bulb. Place the bulb in a 500-ml Erlenmeyer flask containing 100 ml of 95-percent ethanol, and crush the bulb and stem with a heavy glass rod. Wash the rod with 95-percent ethanol and rotate the contents of flask gently to effect solution. Stir 10 grams of sodium peroxide into 100 ml of cold water, and transfer immediately to the flask containing the specimen and ethanol. Swirl to mix the solution and allow to stand, loosely stoppered, for 1/2 hour. Evaporate to approximately 100 ml, cool to room temperature, neutralize with 1 plus 1 nitric acid, add 3 ml excess, and determine total chlorine by the Volhard method. Calculate the percent by weight purity as follows:

$$\text{Percent purity} = \frac{5.48(DE - FG)}{W} - 1.54H$$

where: D = Milliliters of silver nitrate solution used,
E = Normality of the silver nitrate solution,
F = Milliliters of potassium thiocyanate solution used,
G = Normality of the potassium thiocyanate solution,
H = Percent free chlorine determined in (a), and
W = Weight of specimen in grams.

5. PACKAGING

5.1 Preservation, level C. A quantity of chloropicrin as specified (see 6.2) shall be preserved in a clean, dry unit container. Unit containers, unit quantities, and the manner of preservation shall be in compliance with Department of Transportation (DOT) regulations and all other applicable

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regulatory requirements. The preservation shall provide protection against damage and leakage during shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Packs of unit containers shall comply with DOT regulations, Uniform Freight Classification Rules, National Motor Freight Classification Rules, and other carrier regulations applicable to the mode of transportation.

5.3 Unitization. Packs of chloropicrin shall be unitized to meet carrier acceptance and assure safe delivery of the packs at destination.

5.4 Marking. Unit packs, packs, and unitized loads shall be marked in accordance with MIL-STD-129 and shall show date of pack and lot or batch number. In addition, unit containers shall be marked to show the following precautionary information:

HAZARDS: Highly Toxic by Ingestion
Toxic by Inhalation
Irritant Skin and Eye

DANGER! MAY BE FATAL IF SWALLOWED
HARMFUL IF INHALED
CAUSES IRRITATION

Avoid breathing vapor.
Keep container closed.
Use with adequate ventilation.
Avoid contact with eyes, skin and clothing.
Wash thoroughly after handling.

POISON
Call a physician.

FIRST AID: If swallowed, administer large quantities of water and induce vomiting by having patient touch the back of his throat with finger. Never give anything by mouth to an unconscious person. If inhaled, remove to fresh air. If not breathing, give artificial respiration, preferably mouth-to-mouth. If breathing is difficult, give oxygen. In case of contact, immediately flush eyes with plenty of water. Wash clothing before re-use.

6. NOTES

6.1 Intended use. Chloropicrin is intended for use as an odorant additive in methyl bromide insecticide.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Unit quantity required (see 5.1)

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6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Sampling and testing precautions. Chloropicrin is an extremely toxic substance which will cause lachrymation, vomiting, bronchitis, and pulmonary edema. Personnel must exercise caution in sampling and testing chloropicrin to avoid ingestion, inhalation, or contact.

6.5 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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